

BiCare: Bi-Objective Set Similarity with Box Embeddings for Safe and Effective Healthcare Decision Support

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Abstract

Healthcare decision support requires balancing therapeutic effectiveness and medication safety, both of which fundamentally rely on complex set-level relations. While effectiveness involves coverage and overlap between patient needs and medication effects, safety necessitates modeling exclusionary Drug-Drug Interactions (DDIs). Existing vector-based methods, limited to point-level similarities, fail to capture these complex relations. To this end, we propose BiCare, a bi-objective set similarity framework based on box embeddings for safe and effective decision support. BiCare initializes clinical concepts as hyperrectangular boxes to enable expressive representation. Then, BiCare quantifies effectiveness via overlap-based set alignment and models safety using a patient-aware cosine compatibility that captures DDIs by assessing directional inconsistencies between medication and patient boxes. A dynamic weight averaging strategy further enables adaptive bi-objective balancing. Experiments on two real-world datasets demonstrate BiCare's state-of-the-art effectiveness with competitive safety.

CCS Concepts

• Applied computing → Health informatics.

Keywords

Clinical Decision Support, Box Embeddings, Bi-Objective Learning, Set Similarity, Drug-Drug Interactions, Healthcare

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1 Introduction

Healthcare decision support systems play a crucial role in clinical workflows by helping clinicians select appropriate interventions while balancing two competing objectives: therapeutic effectiveness [13] and medication safety [6]. Real-world Electronic Health Records (EHRs) are inherently set-structured, where each patient

visit is represented as a set of diagnosed diseases, clinical findings, and prescribed medications. Consequently, bi-objective healthcare decision-making on such data fundamentally relies on set-level relations: effectiveness depends on coverage and partial overlap between patient needs and medication effects [1, 14], while medication safety requires modeling exclusionary Drug-Drug Interactions (DDIs) among co-prescribed medications [9, 11].

Existing bi-objective approaches typically embed clinical concepts as vectors, enabling only point-level similarity computation [10, 15]. Such vector-based models struggle to express more complex set relations (e.g., containment, partial overlap, and exclusionary DDI) that are crucial for bi-objective healthcare decision support. Recent advances in box embeddings [8] provide a promising solution by encoding concepts as hyperrectangles, which enable expressive modeling for set-level similarity computation. Yet, leveraging such box embeddings for bi-objective decision remains non-trivial, as therapeutic effectiveness and medication safety correspond to fundamentally different geometric interactions, which is an unaddressed challenge in existing box-based works.

In this paper, we propose BiCare, a bi-objective set similarity framework based on box embeddings for safe and effective healthcare decision support. BiCare represents diagnoses, procedures, and medications as boxes in a shared embedding space and constructs patient representations via set-based aggregation. For therapeutic effectiveness, BiCare employs overlap-based similarity that quantifies the proportion of a medication's therapeutic scope that aligns with patient needs, naturally capturing set-level coverage relations. For medication safety, BiCare introduces cosine-based compatibility that captures directional alignment, where two medications may both overlap with patient needs yet exhibit low cosine similarity (large angular distance), signaling potential interactions. Furthermore, BiCare employs Dynamic Weight Averaging (DWA) [12] for adaptive bi-objective balancing. Extensive experiments on real-world EHR datasets validate the effectiveness of BiCare, demonstrating its superior Jaccard performance compared to state-of-the-art methods, while maintaining a comparable DDI rate.

2 The BiCare Framework

2.1 Preliminaries

Given a patient p 's EHR data represented as a sequence of visits $\mathcal{V}_p = \{v_1, v_2, \dots, v_t\}$, where each visit v_t at time t contains a set of diagnoses $\mathcal{D}_t$, procedures $\mathcal{C}_t$, and medications $\mathcal{M}_t$. Our goal is to support clinical decision-making by optimizing two objectives: (1) Therapeutic effectiveness: identify candidate medications $\mathcal{M}_{\text{cand}}$ that effectively address the patient's current clinical needs, and

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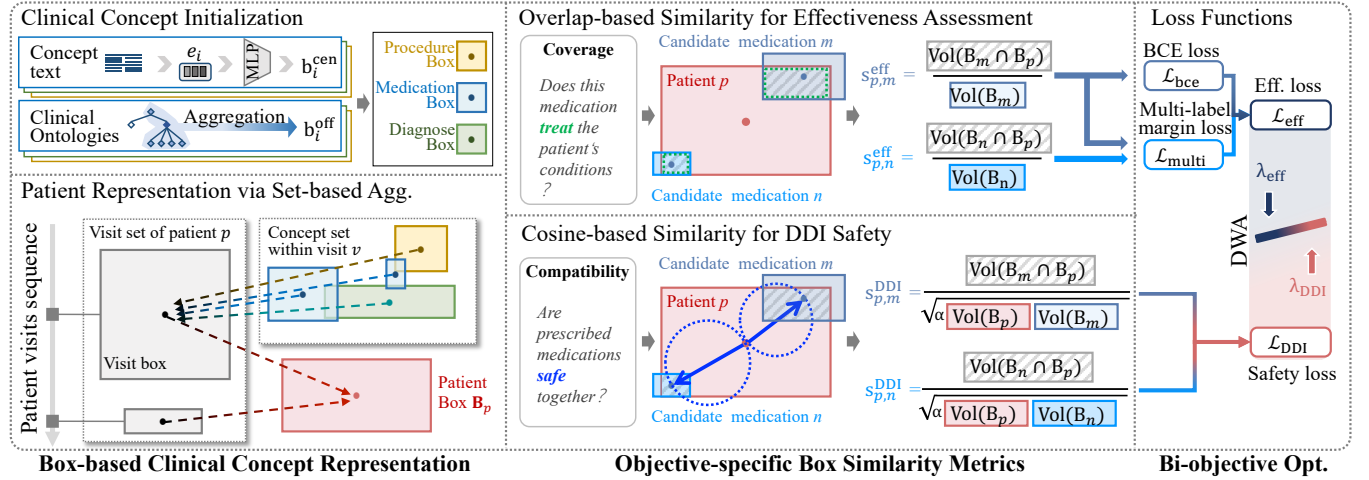


Figure 1: The overall framework of our proposed BiCare.

(2) Medication safety: ensure $\mathcal{M}_{\text{cand}}$ do not cause harmful DDIs. The overall Framework of BiCare is shown in Figure 1.

2.2 Box-based Clinical Concept Representation

2.2.1 Clinical Concept Initialization. Each clinical concept i (diagnosis d , procedure c , or medication m) is encoded as a dim -dimensional hyperrectangle. Each box B_i is defined by its center b_i^{cen} and offset b_i^{off} :

$$B_i = (b_i^{\text{cen}}, b_i^{\text{off}}), \quad b_i^{\text{min}} = b_i^{\text{cen}} - b_i^{\text{off}}, \quad b_i^{\text{max}} = b_i^{\text{cen}} + b_i^{\text{off}}, \quad (1)$$

where b_i^{min} and b_i^{max} are the minimum and maximum corners. The box centers are initialized from BioBERT [4] embeddings e_i via a learnable Multi-Layer Perceptron (MLP), and box offsets are refined via graph convolution on hierarchical ontologies:

$$b_i^{\text{cen}} = \text{MLP}(e_i), \quad b_i^{\text{off}} = \text{AGG}\left(\left\{\phi(b_j^{\text{cen}}, W_r) \mid (j, r) \in \mathcal{N}(i)\right\}\right), \quad (2)$$

where $\mathcal{N}(i)$ denotes neighbors of i , $\phi(\cdot, W_r)$ is a composition function with learnable weights, and $\text{AGG}(\cdot)$ aggregates neighbors.

2.2.2 Patient Representation via Set-based Aggregation. A patient's clinical profile evolves across multiple visits, each involving multiple concepts. We construct a unified patient box $B_p = (b_S^{\text{cen}}, b_S^{\text{off}})$ by hierarchically aggregating sets of concept boxes. For a set S of clinical entities (either concepts within a visit v_t , or visits within patient history $\mathcal{V}_p$), the box center is computed via weighted aggregation:

$$b_S^{\text{cen}} = \sum_{i \in S} w_i \cdot b_i^{\text{cen}}, \quad w_i = \frac{\exp(f(\cdot, \theta))}{\sum_{j \in S} \exp(f(j, \theta))}, \quad (3)$$

where f is context-specific weighting function: $f(\cdot, \theta) = \text{MLP}_\alpha(b_i^{\text{cen}})$ for concept-level aggregation, and $\text{MLP}_T(\Delta t)$ for temporal visit aggregation, Δt is the time interval to the current time. The aggregated offset preserves the broadest coverage via max-pooling:

$$b_S^{\text{off}} = \max\{b_i^{\text{off}} \mid i \in S\}. \quad (4)$$

2.3 Objective-specific Box Similarity Metrics

BiCare employs distinct similarity metrics for effectiveness and safety, reflecting that effectiveness requires measuring coverage while safety requires evaluating compatibility. We first define fundamental geometric operations. The volume of a box B_i quantifies

the extent of a concept in latent space:

$$\text{Vol}(B_i) = \prod_{k=1}^{\text{dim}} \max(b_{i,k}^{\text{max}} - b_{i,k}^{\text{min}}, 0). \quad (5)$$

Then, the intersection volume of two boxes is calculated as:

$$\text{Vol}(B_i \cap B_j) = \prod_{k=1}^{\text{dim}} \max(\min(b_{i,k}^{\text{max}}, b_{j,k}^{\text{max}}) - \max(b_{i,k}^{\text{min}}, b_{j,k}^{\text{min}}), 0). \quad (6)$$

Overlap-Based Similarity for Effectiveness Assessment.

For therapeutic effectiveness, we employ an asymmetric overlap metric that quantifies the fraction of patient p 's clinical needs with medication m 's therapeutic space, calculated as:

$$s_{p,m}^{\text{eff}} = \frac{\text{Vol}(B_p \cap B_m)}{\text{Vol}(B_m)}. \quad (7)$$

Cosine-Based Similarity for DDI Safety. For medication safety, we introduce a cosine-based compatibility that implicitly encodes DDI constraints through angular positioning. BiCare adopts a factored design where each medication box B_m receives an individual compatibility score with the patient box B_p , calculated as:

$$s_{p,m}^{\text{DDI}} = \frac{\text{Vol}(B_p \cap B_m)}{\sqrt{\alpha \cdot \text{Vol}(B_p) \cdot \text{Vol}(B_m)}}, \quad (8)$$

where $\alpha = \langle \text{Vol}(B_m) \rangle / \langle \text{Vol}(B_p) \rangle$ is an adaptive normalization factor computed as the ratio of mean medication box volume to mean patient box volume over the training set, ensuring numerical stability and scale-invariance. The square root normalization in the denominator, analogous to cosine similarity's normalization by vector magnitudes, balances the influence of patient and medication box sizes while preserving their individual contributions. Unlike overlap, which measures absolute box intersection volume, cosine-based similarity captures directional alignment. Medications may individually overlap with a patient (high effectiveness) but point in conflicting directions (low cosine), signaling potential interactions.

2.4 Bi-objective Optimization

BiCare optimizes therapeutic effectiveness and medication safety jointly through a composite loss function with adaptive weighting.

Loss Functions. The effectiveness loss combines binary cross-entropy (BCE) and multi-label margin losses:

$$\mathcal{L}_{\text{eff}} = \lambda_r \cdot \mathcal{L}_{\text{bce}} + (1 - \lambda_r) \cdot \mathcal{L}_{\text{multi}}, \quad (9)$$

where $\mathcal{L}_{bce}$ measures prediction accuracy, $\mathcal{L}_{multi}$ enforces separation margins between prescribed and non-prescribed medications, and λ_r is a balancing hyperparameter. Specifically:

$$\mathcal{L}_{bce} = - \sum_{p \in \mathcal{P}} \sum_{m \in \mathcal{M}} [y_{p,m} \ln(s_{p,m}^{\text{eff}}) + (1 - y_{p,m}) \ln(1 - s_{p,m}^{\text{eff}})], \quad (10)$$

$$\mathcal{L}_{multi} = \sum_{p \in \mathcal{P}} \sum_{\substack{y_{p,m}=1, y_{p,n}=0 \\ m, n}} \frac{\max\left(0, 1 - \left(s_{p,m}^{\text{eff}} - s_{p,n}^{\text{eff}}\right)\right)}{|\mathcal{M}|}, \quad (11)$$

where $y_{p,m} = 1$ or $y_{p,n} = 0$ indicates whether medication is prescribed for patient p in the ground truth.

The DDI safety loss penalizes co-prescription of interacting medication pairs:

$$\mathcal{L}_{DDI} = \sum_{p \in \mathcal{P}} \sum_{m \in \mathcal{M}} \sum_{n \in \mathcal{M}} [s_{p,m}^{\text{DDI}} \cdot s_{p,n}^{\text{DDI}} \cdot D_{mn}], \quad (12)$$

where D_{mn} is the DDI severity between medications m and n from the TWOSIDES database [9]. During training with DDI loss, the model learns to position interacting drug boxes such that they exhibit low cosine-based compatibility scores, even when each drug individually matches the patient well.

Dynamic Weight Averaging. To avoid manual tuning and ensure balanced learning across effectiveness and safety objectives, BiCare employs DWA [12], which adaptively adjusts loss weights based on each objective’s learning progress. At training iteration a , the weight for objective $z \in \{\text{eff}, \text{DDI}\}$ is:

$$\lambda_z(a) = \frac{2 \cdot \exp(w_z(a-1)/\tau)}{\sum_{z'} \exp(w_{z'}(a-1)/\tau)}, \quad w_z(a) = \frac{\mathcal{L}_z(a)}{\mathcal{L}_z(a-1)}, \quad (13)$$

where $w_z(a)$ measures the relative descent rate, and τ is a temperature parameter. DWA automatically increases the weight of slower-converging objectives. The final loss is:

$$\mathcal{L}_{\text{final}} = \lambda_{\text{eff}}(a) \cdot \mathcal{L}_{\text{eff}} + \lambda_{\text{DDI}}(a) \cdot \mathcal{L}_{\text{DDI}}. \quad (14)$$

3 Experiment

3.1 Experiment Settings

Dataset and Metric. We use two real-world EHR datasets. **MIMIC-III** [3] is an open-access database including anonymized records of more than 40,000 patients. **MIMIC-IV** [2] contains clinical data on over 65,000 ICU and 200,000 emergency department patients. We choose patients with at least two visits, and filter out the patients without valid clinical concept codes for each visit. We utilize **Jaccard** metrics for effectiveness and **DDI rate** metrics for safety evaluation, following previous works [11, 15]. **# Med** captures the average number of prescribed medications. **Avg. Rank** report the average rank over four metrics, i.e., Jaccard/DDI on MIMIC-III/IV.

Baselines. We adopt 10 baselines from two perspectives: (1) *Effectiveness-focused methods*: **ECC** [5], **LEAP** [13]; **BioBERT** [4], **OntoMedRec** [7]. **UDC** [14]. **RETAIN** [1], (2) *Bi-objective methods*: **GAMENet** [6], **SafeDrug** [10], **MoleRec** [11], **DEPOT** [15].

Implementation Details. Both datasets are split into train/validation/test sets (4:1:1), following [6, 10, 11, 15]. To simulate real-world sparse data scenarios, we sampled 5% of the training set and conducted multiple tests. Baselines are optimized using the standard Adam optimizer, with hyperparameters meticulously tuned following the recommendations in original papers. The dimension

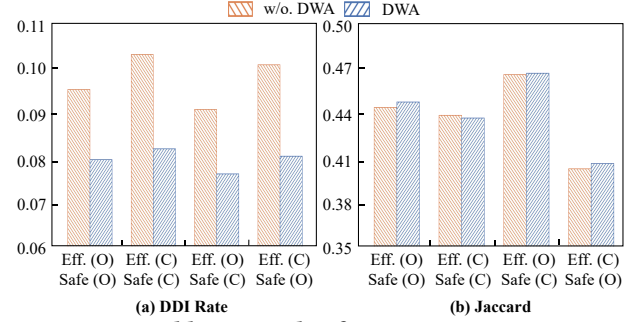


Figure 2: Ablation study of BiCare on MIMIC-III.

$\dim$ is 16, batch size is 128, balancing coefficient λ_r is 0.93, and temperature parameter in DWA τ is 1 unless otherwise specified.

3.2 Overall Performance Comparison

Experimental results are shown in Table 1, BiCare consistently achieves the highest effectiveness while maintaining competitive safety, demonstrating successful bi-objective optimization.

Superiority in Effectiveness. BiCare substantially outperforms all effectiveness-focused baselines in therapeutic accuracy. Compared to the most effective baseline UDC, BiCare represents an average improvement of 1.7%. These gains are even more pronounced against the Bi-objective baselines, showing approximately 4% improvement over DEPOT on MIMIC-III. The improvement stems from our overlap-based geometric similarity, which enables precise alignment between patient boxes and medication boxes, capturing nuanced condition–treatment relationships.

Competitive Safety Performance. While maintaining effectiveness, BiCare demonstrates competitive DDI rates of 0.0767 on MIMIC-III and 0.0727 on MIMIC-IV, only marginally higher than the most safety-focused baseline SafeDrug, yet substantially lower than other baselines. Importantly, BiCare’s safety is achieved without sacrificing effectiveness. The combination of top-tier effectiveness and competitive safety yields the overall best ranking (1.50), confirming the advantage of BiCare’s cosine-based compatibility metric and dynamic bi-objective weighting, which successfully balances bi-objectives for safe and effective healthcare.

3.3 Ablation Study

To assess the contribution of each component in BiCare, we conduct ablation experiments on MIMIC-III (Figure 2). We vary the similarity for effectiveness (Eff.) and DDI safety (Safe), where “O” denotes the overlap similarity and “C” denotes the cosine compatibility, each further evaluated with and w/o. DWA.

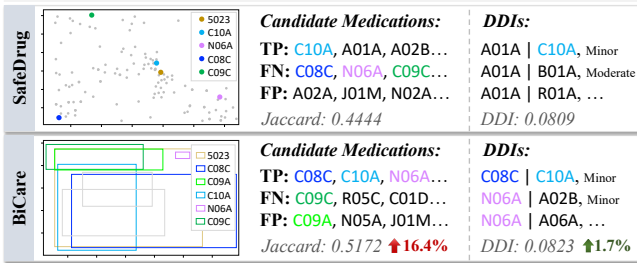
Impact of DWA. Without DWA module, the optimization exhibits a strong bias at the substantial cost of elevated DDI rates. Introducing DWA alleviates this imbalance, reducing DDI in Eff. (O) $\times$ Safe (C) effectively (about 16% relative decrease) while keeping Jaccard virtually unchanged. These results demonstrate the critical role of DWA in adaptive bi-objective balancing.

Objective-Specific Similarity Design. The ablation results also provide strong evidence for employing distinct similarity metrics for different objectives. When fixing the safety metric and DWA setting, using overlap for effectiveness (Eff.(O)) consistently outperforms cosine (Eff.(C)), yielding an average 8.6% higher Jaccard and 6.4%

Table 1: Experimental results. The best results are highlighted in bold while the second best are marked with underline.

Method	DDI ↓	MIMIC-III Jaccard ↑	# Med	DDI ↓	MIMIC-IV Jaccard ↑	# Med	Avg. Rank ↓
ECC	0.0898±0.0006	0.4087±0.0025	13.4845±0.0589	0.0937±0.0008	0.3818±0.0019	8.2940±0.0724	8.50
LEAP	0.0875±0.0003	0.3939±0.0024	18.5664±0.0398	0.0752±0.0005	0.3539±0.0026	8.9526±0.0269	7.75
BioBERT	0.1094±0.0004	0.4263±0.0022	25.5256±0.0637	0.0994±0.0008	0.4108±0.0022	15.5734±0.0637	8.50
OntoMedRec	0.0785±0.0003	0.4285±0.0021	15.8046±0.1548	0.0765±0.0006	0.4117±0.0023	11.6739±0.0924	5.00
UDC	0.1060±0.0003	0.4631±0.0023	16.3416±0.1096	0.1127±0.0002	0.4354±0.0024	11.1518±0.0933	6.50
RETAIN	0.0897±0.0008	0.3814±0.0028	12.7079±0.0564	0.0985±0.0012	0.3373±0.0007	6.7078±0.0845	9.75
GAMENet	0.0805±0.0003	0.4105±0.0023	15.3363±0.0518	0.0977±0.0014	0.4099±0.0039	10.5075±0.0936	7.00
SafeDrug	0.0700±0.0002	0.4317±0.0026	16.9059±0.0926	0.0631±0.0003	0.4083±0.0030	12.9756±0.0965	3.75
MoleRec	0.0817±0.0002	0.4459±0.0020	22.6801±0.1565	0.0774±0.0008	0.4252±0.0027	12.6647±0.0953	5.00
DEPOT	0.0769±0.0003	0.4530±0.0028	18.9076±0.1078	0.0736±0.0003	0.4348±0.0008	13.5496±0.0601	2.75
BiCare	<u>0.0767±0.0002</u>	0.4709±0.0020	21.8419±0.0988	<u>0.0727±0.0003</u>	0.4428±0.0022	13.9161±0.0904	1.50

Patient:5023,visit:1,diagnoses: congestive heart failure, hypertensive chronic kidney disease, depressive...

**Figure 3: Interpretable case study on MIMIC-III.**

lower DDI. Conversely, when fixing the effectiveness metric, using cosine for safety (Safe(C)) outperforms overlap (Safe(O)), achieving 6.7% lower DDI and 1.1% higher Jaccard on average.

3.4 Interpretable Case Studies

We showcase BiCare’s interpretability via a MIMIC-III case (Figure 3), where it trades marginally lower safety (DDI +1.7%) for significantly higher effectiveness (Jaccard +16.4%) compared to the safest baseline SafeDrug. This stems from their geometric reasoning. For example, to avoid DDI, SafeDrug’s point vectors unduly separate C08C: Amlodipine(FN) from C10A: Atorvastatin(TP), risking under-treatment for heart failure. In contrast, BiCare’s boxes maintain high overlap with the patient box for both medications, and its DWA strategy balances this therapeutic overlap against their angular distance (a clinically manageable, common DDI).

4 Conclusion

This work introduced BiCare, a box-embedding approach tailored for set-level bi-objective clinical decision support. Through overlap-based effectiveness modeling, cosine-driven compatibility for DDI safety, and adaptive weighting, BiCare achieves state-of-the-art effectiveness with competitive safety on two real-world datasets. Future work may explore incorporating additional clinical constraints and extending to broader multi-objective healthcare tasks.

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